Disagreement-Aware Probabilistic U-Net:
Making Predictive Diversity Land Where Annotators Actually Disagree

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August 19, 2026

Selected paper. S. A. A. Kohl et al., *A Probabilistic U-Net for Segmentation of Ambiguous Images*, NeurIPS 2018 [1].

Proposed extension. We supervise *where* human annotators disagree. A single-convolution head predicts the inter-rater disagreement map from the image alone, and an alignment loss drives the model's own sample diversity into those same pixels.

1 Introduction

Semantic segmentation assigns a label to every pixel of an image, and training such a model normally assumes that each image comes with one correct mask. In medical imaging that assumption rarely holds. Asked to outline the same lung nodule, several qualified radiologists produce visibly different contours, and they sometimes disagree about whether there is a nodule there at all. A conventional U-Net [4] trained on one expert's mask, or on the average of several, returns a single confident segmentation and discards that disagreement exactly where it is most clinically relevant.

The Probabilistic U-Net [1] addresses this by pairing a U-Net with a conditional variational autoencoder. Instead of a single output, the model learns a low-dimensional latent space in which each point corresponds to a plausible reading of the image; drawing several latents and decoding them produces several segmentations. Which region of that space a given expert's reading occupies is taught by a second encoder, the posterior, which during training sees the mask as well as the image. At test time no mask is available, so the latents are drawn from the prior encoder, which sees the image alone. Each sample is a coherent whole mask rather than a map of independent per-pixel probabilities, which is the paper's central argument: a per-pixel map cannot express which pixels vary *together*, so it has no way to represent two distinct but internally consistent readings of the same scan.

Scoring such a model needs a metric that compares two *sets* of masks rather than two masks, and the paper uses the generalized energy distance. For one image it takes the set of masks the model sampled and the set of four expert masks, measures how far apart individual masks are using their overlap, and returns a single number that is small when the two sets resemble one another.

We chose this paper because of a question it forced us to ask: should a model always commit to its single most likely output? When the experts themselves disagree there is no one right mask to choose, and a single confident answer hides the ambiguity rather than reporting it. We liked that the paper keeps the disagreement instead of averaging it away, and we wanted to push the idea one step further.

That single number is where we saw an opening. It records how *much* the sampled masks vary, but nothing about *where* in the image they vary, so a model could place the right amount of variability in entirely the wrong region and score just as well as one that places it correctly. Since LIDC-IDRI [3] provides four independent expert masks per image, that location can be measured directly, so we use it as a supervision signal. The two additions we make are both small, and they fare differently. Predicting where the experts will disagree works well. Forcing the model's own variability into those same pixels helps only a little, for a reason that lies in the shape of the signal four annotations can give.

2 Proposed extension

The Probabilistic U-Net supervises its diversity only indirectly. At each training step the posterior network is shown one randomly chosen expert, so over many steps the model does see the full range of annotations, but no single update ever tells it which parts of the image the experts actually argued about. The evaluation shares that blind spot. Neither the objective nor the metric separates a model whose variability sits on the disputed boundary from one that varies somewhere else entirely.

Two things make this worth addressing. The information is already in the data and goes unused: four masks per image give a per-pixel picture of where the experts disagreed, at no extra annotation cost. And a model that can point at the disputed region is more useful as a second reader than one reporting a single overall confidence, which is the clinical motivation of the original paper. The change also fits the existing architecture rather than replacing it: both additions attach to the last U-Net feature map, leaving the encoder, decoder, latent space and original loss untouched, so any difference we measure follows from the added supervision rather than from added capacity.

We add two components. The **disagreement head** predicts, for every pixel, how strongly the four experts disagreed there. It sees only the image, never a sample, so at inference it can flag the contested region without any annotation to compare against. The **alignment loss** takes the spread of the model's own samples and penalises it wherever that spread does not coincide with the measured expert disagreement. Both add pixel-level information answering the "where" question while leaving intact the original model's view of a segmentation as one coherent mask.

This gives two claims to test. **Q1:** where annotators will disagree is predictable from the image alone, so the head should approximate the measured disagreement and beat a trivial predictor that only marks lesion boundaries. **Q2:** adding the alignment loss should improve the agreement between the model's uncertainty and the experts' disagreement, without hurting segmentation quality or the energy distance.

We considered two alternatives. Conditioning the prior on the predicted disagreement was rejected because that prediction is itself a deterministic function of the image, so it adds no new information and would blur the comparison we set out to make. Per-rater output heads or annotator confusion matrices [5] were rejected because the LIDC release does not record which radiologist produced which mask, and the four raters differ from scan to scan, so a per-rater parameterisation would have nothing stable to learn.

3 Methodology

3.1 Model

We reimplemented the Probabilistic U-Net in PyTorch, since the official release is an older TensorFlow codebase that no longer runs on Colab. The model has three parts. The U-Net is a standard encoder–decoder with skip connections: the encoder halves the spatial resolution four times while doubling the channel count, starting from 32 channels, and the decoder mirrors it, concatenating the matching encoder feature map at every step. Alongside it are two small convolutional encoders that each output the mean and variance of a six-dimensional Gaussian. The prior encoder sees only the image; the posterior encoder, used during training, sees the image together with one expert's mask. A sampled latent z is broadcast into a six-channel spatial map, concatenated onto the U-Net's last feature map, and passed through three 1×1 convolutions (the `fcomb`) that mix the two and produce the class logits. Only this last block has to be recomputed to draw another sample, which is what makes sampling cheap. The disagreement head branches off the same last feature map one step earlier, upstream of the concatenation, and is a single 1×1 convolution followed by a sigmoid. Figure 1 shows the arrangement.

3.2 Disagreement maps and the two new loss terms

Three quantities have to be kept distinct. The first, D^{GT}, is the *human* disagreement, computed from the four masks alone. With $P_i = \frac{1}{G}\sum_g Y_i^{(g)}$ the per-pixel mean over the $G = 4$ expert masks $Y^{(g)}$, we take its normalised binary entropy,

$$D_i^{\mathrm{GT}} = -\frac{P_i \log(P_i + \epsilon) + (1 - P_i)\log(1 - P_i + \epsilon)}{\log 2} \in [0, 1],$$

which is 0 wherever all four experts agree and 1 where they split two against two. The second, $\hat{D}$, is the head's *prediction* of D^{GT}. The third, U, is the *model* uncertainty: we draw K latents from the prior, decode each into a foreground-probability map $Q^{(k)}$, average them into $\bar{Q}_i = \frac{1}{K}\sum_k Q_i^{(k)}$ and apply the same entropy,

$$U_i = -\frac{\bar{Q}_i \log(\bar{Q}_i + \epsilon) + (1 - \bar{Q}_i)\log(1 - \bar{Q}_i + \epsilon)}{\log 2}.$$

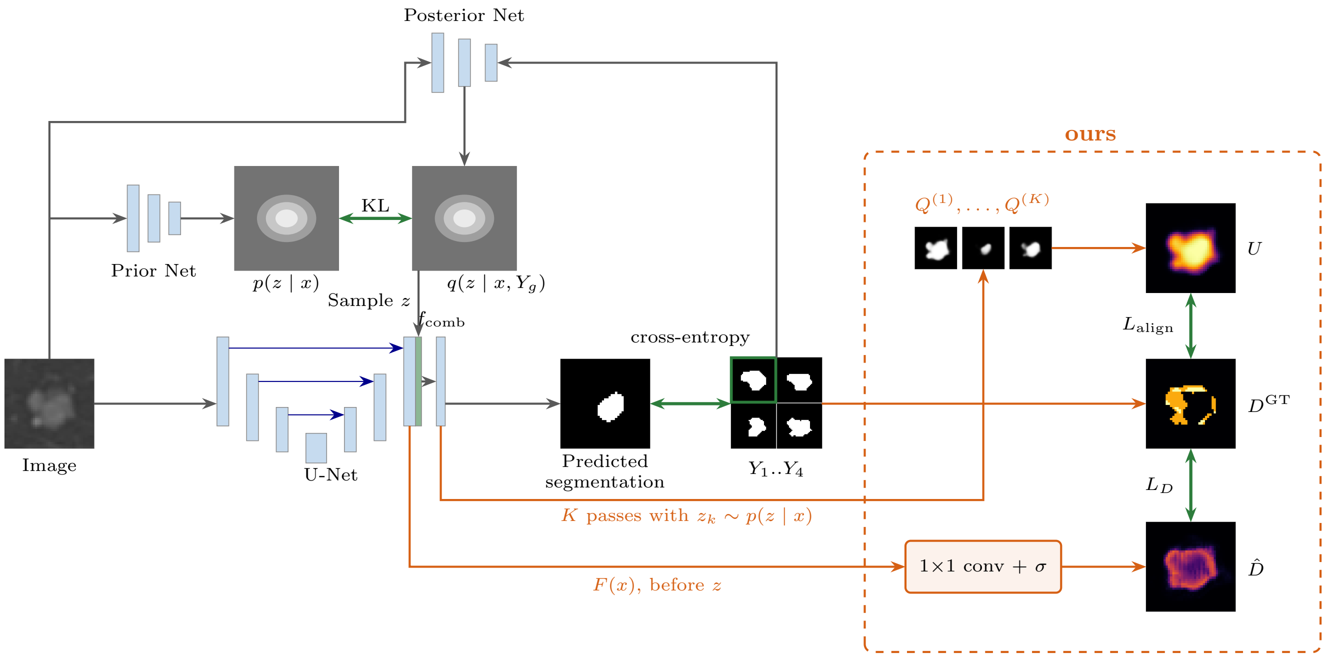

Figure 1: **The disagreement-aware Probabilistic U-Net (training).** Outside the dashed box, the original model: the posterior sees the image and *one* randomly drawn expert Y_g (green outline), and the sampled z (green block, 6 channels) is concatenated to the last U-Net activation (32 channels). The three $1{\times}1$ convolutions of f_{comb} combine them. Inside the box, our extension. Note that $F(x)$ is taken *before* z is added, so $\hat{D}$ depends on the image alone, and that the K passes behind U each draw a fresh z_k from the *prior*. **Every thumbnail is real**, from test crop `LIDC-IDRI-0217` decoded by our trained full model and zoomed to the nodule; D^{GT}, U and $\hat{D}$ share one $[0, 1]$ colour scale, so they can be compared directly. The three $Q^{(k)}$ shown are non-empty draws: 8 of the 16 samples for this crop are empty, though all four experts segmented the nodule.

Using one functional for both means U and D^{GT} differ only in what is averaged, model samples in one case and expert votes in the other, so any discrepancy between them is a discrepancy of location on a common $[0, 1]$ scale.

The head is trained with $L_D = \text{MSE}(\hat{D}, D^{\text{GT}})$ and the alignment loss is the mean absolute difference between the two maps, $L_{\text{align}} = \frac{1}{HW} \sum_i |U_i - D_i^{\text{GT}}|$, giving the full objective

$$L = L_{\text{PU-Net}} + \lambda_D L_D + \lambda_A L_{\text{align}},$$

where $L_{\text{PU-Net}}$ is the original evidence lower bound: a cross-entropy reconstruction of the sampled expert's mask, summed over pixels, plus β times the Kullback–Leibler divergence between posterior and prior.

3.3 Data

We use the LIDC-IDRI 2D crops from DeepMind's preprocessed release (CC BY 3.0), published with the Hierarchical Probabilistic U-Net [2], whose preprocessing follows Appendix H.1 of the original paper. The splits contain 8843 training, 1993 validation and 1980 test images from 530, 111 and 103 patients respectively; images are 180×180 8-bit grayscale with four masks each. We take a random 128×128 crop, scale images to $[0, 1]$ and binarise the masks. About 65% of training crops contain at least one empty mask, which reflects disagreement about whether a lesion exists at all rather than about where its boundary lies, and this shapes how the metrics must be computed.

3.4 Evaluation

Dice and IoU check that the extension does not damage segmentation quality. For two binary masks A and B they are $\text{Dice}(A, B) = 2|A \cap B|/(|A| + |B|)$ and $\text{IoU}(A, B) = |A \cap B|/|A \cup B|$, and we compute them between the model's deterministic prediction and each of the four masks in turn. The distribution as a whole is scored by the generalized energy distance. Writing S, S' for independent model samples and Y, Y' for independent expert masks,

$$D^2_{\mathrm{GED}} = 2\,\mathrm{E}[d(S,Y)] - \underbrace{\mathrm{E}[d(S,S')]}_{\text{sample diversity}} - \mathrm{E}[d(Y,Y')]\,, \qquad d = 1 - \mathrm{IoU}.$$

Both IoU and Dice are 0/0 when the two masks being compared are empty, so a convention is needed for those pairs. For the energy distance we follow the paper's Appendix B, which sets $d \equiv 0$, so that agreeing there is no lesion counts as agreement. The paper does not say what to do for Dice, and there we made our own choice: we drop empty-versus-empty pairs from the average rather than scoring them zero, since scoring them zero would penalise every model equally on the 65% of crops that contain an empty mask and would hide the differences we care about.

Our primary measure is the correlation $\mathrm{corr}(U, D^{\mathrm{GT}})$, which asks whether the model is uncertain in the same places the humans disagree, and correspondingly $\mathrm{corr}(\hat{D}, D^{\mathrm{GT}})$ for the head. Because disagreement in LIDC concentrates on lesion boundaries, a map that merely marks a boundary already correlates with D^{GT}, so the head is judged against a trivial *boundary-band control*: a 5-pixel band around the edge of the model's own predicted mask. The control uses no expert information, so unlike a band drawn around the ground truth it is something a deployed model could actually compute, and the head has to beat it to have learned anything about the annotators. We also report the $\mathrm{E}[d(S,S')]$ term of the energy distance on its own, since it detects samples that have collapsed onto a single mask, together with *nestedness*: the fraction of images for which, after sorting that image's samples by area, each one is wholly contained in the next. Nestedness catches a subtler failure, where the model only inflates and shrinks one shape without ever proposing a different one, so its samples can look varied while really having a single degree of freedom.

All code is written in Python with PyTorch, NumPy and Matplotlib. The submitted implementation runs in Google Colab as a self-contained notebook that executes the whole pipeline in a reduced configuration for a quick demonstration; the same code, run at full length, produced every number reported below.

3.5 Training and ablation design

We follow the original paper's training recipe: $\beta = 1$ with a six-dimensional latent, Adam with learning rate 10^{-4} decayed to 10^{-6} in five steps, batch size 32 and weight decay 10^{-5}. Holding these fixed is deliberate, since it leaves the loss terms as the only thing that differs between the three models we compare, all of which come from one codebase.

We depart from the paper in a few places. Training runs for 100k steps rather than 240k, limited by the compute available to us, so our absolute numbers sit above the published ones while comparisons among our own models remain fair. Augmentation is the random translated crop only, where the paper also uses elastic deformation, rotation, shearing and scaling. The two new loss weights, $\lambda_D = 0.01$ and $\lambda_A = 10^{-5}$, come from short 10k-step trials: larger values bought very little and cost a great deal, with $\lambda_A = 10^{-3}$ improving the uncertainty correlation by roughly 0.02 while Dice fell from 0.30 to 0.12, so we kept the smallest weights that still had a measurable effect. We clip the gradient norm and clamp the latent log-variance to keep training stable. Each step draws $K = 4$ samples, and evaluation uses 16 per image. The three arms add one component at a time:

Arm	Disagreement head	Alignment loss	Objective
A – baseline PU-Net	no	no	$L_{\text{PU-Net}}$
B – + head	yes	no	$+\lambda_D L_D$
C – full model	yes	yes	$+\lambda_D L_D + \lambda_A L_{\text{align}}$

4 Results and analysis

4.1 Quantitative results

Table 1 reports the held-out test split (1980 images, 16 samples per image), with every arm read from its 100k-step checkpoint.

Q1 is borne out. The head reaches $\mathrm{corr}(\hat{D}, D^{\mathrm{GT}}) = 0.627$ against its own boundary-band control of 0.425, a margin of +0.20, and both arms carrying a head behave the same way. A single convolution therefore recovers something about annotator behaviour that the outline of the model's own prediction does not.

Arm	Dice ↑	IoU ↑	GED ↓	$\mathrm{corr}(U, D^{\mathrm{GT}})$ ↑	$\mathrm{corr}(\hat{D}, D^{\mathrm{GT}})$ ↑	control	$\mathrm{E}[d(S, S')]$	nested ↓
A – baseline	**0.355**	**0.280**	0.394	0.483	—	0.417	0.686	1%
B – + head	0.308	0.241	0.403	0.492	0.609	0.451	0.687	3%
C – full	0.340	0.267	**0.388**	**0.509**	**0.627**	0.425	0.692	1%

Table 1: Test-set results. *control* is the boundary-band predictor for the same model, so $\mathrm{corr}(\hat{D}, D^{\mathrm{GT}})$ should be read against the value beside it.

Q2 goes the right way, but the effect is small. $\mathrm{corr}(U, D^{\mathrm{GT}})$ rises from 0.483 to 0.509 between arms A and C, and arm C simultaneously reaches the best energy distance of the three (0.388) for a 4% relative drop in Dice. The size of the gain is a property of the loss rather than of its weight: the alignment term compares two per-pixel maps, so two sample distributions with the same per-pixel statistics are indistinguishable to it. It can move variance around the image, but it cannot change which masks the model proposes.

4.2 Qualitative results

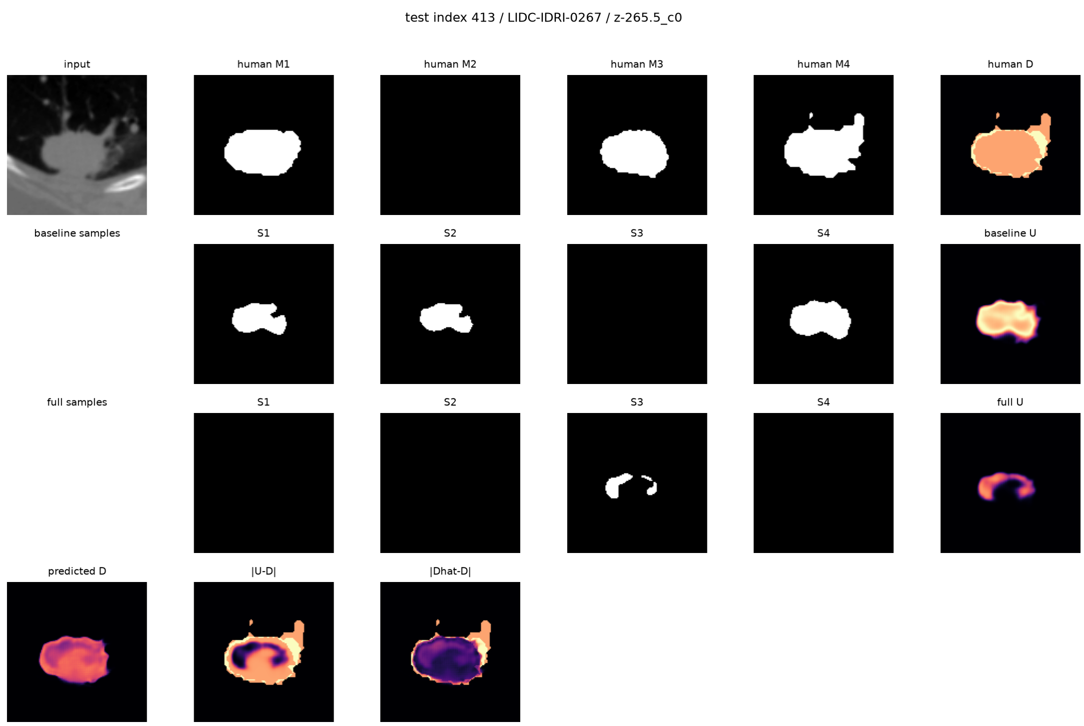

Figure 2: A case where the experts disagree about whether a lesion exists at all (test index 413, LIDC-IDRI-0267). Row 1 shows the CT crop, the four expert masks, of which M2 is empty, and D^{GT}, which covers the whole nodule rather than a rim because the disagreement is about existence rather than boundary placement. Row 2 shows four segmentations sampled from the baseline and the uncertainty map U they produce; row 3 shows the same for our full model. Both reproduce the ambiguity, in that some of their samples contain the nodule and some are empty. Row 4 shows the predicted $\hat{D}$ and the two error maps. $|\hat{D} - D^{\mathrm{GT}}|$ is dark over the lesion while $|U - D^{\mathrm{GT}}|$ is bright: the head recovers this kind of disagreement, but U would have to be maximal across the entire nodule to match D^{GT}, which pulls directly against the segmentation loss.

Figure 2 also exposes a limit of the idea itself. Much of the disagreement in LIDC is about whether a nodule is present at all. Matching U to D^{GT} therefore asks the model to be maximally uncertain across whole lesions, which pulls against the reconstruction term and limits how far the alignment loss can be pushed.

4.3 Challenges encountered

Three problems shaped the final setup. The `fcomb` originally applied no nonlinearity between its 1×1 convolutions, making the block a single affine map; since z enters as a spatially constant vector, every sample was then a scaled version of one fixed shape, and all test images had strictly nested samples. With the activations in place nestedness drops to the 1–3% of Table 1, and only then does the alignment loss have anything to act on.

Our first full-length run collapsed and was discarded. We had averaged the reconstruction cross-entropy over pixels where the original paper sums it, which at 128×128 under-weights the reconstruction by a factor of about 16000 against the KL term, so $\beta = 1$ behaved like $\beta \approx 16000$. The KL reached zero by step 31k and the validation energy distance rose from 0.34 to 0.61 while Dice kept improving, the signature of a network that has become deterministic. Summing the cross-entropy resolved it.

Checkpoint selection caused a subtler problem. Choosing the checkpoint with the best validation energy distance selects a very early step for arms A and B, because a barely-trained model with a wide, uninformative prior scores well on that metric. We therefore report all arms at the same 100k steps.

5 Conclusion

We extended the Probabilistic U-Net so that it reports not only how much its samples vary but where, and the two halves of that came out differently. Reading the disputed region off the image is easy: one convolution, supervised by annotations we already had, beats a boundary-only predictor by 0.20 in correlation. Persuading the model's own samples to vary there is hard, and the 0.026 we gained is close to what this formulation allows, because a loss comparing two per-pixel maps can move variance around an image but cannot change which masks the model proposes. The contribution is therefore not a higher Dice, which the baseline already achieves; it is that the disagreement becomes an output one can look at, and that the samples drift, modestly, towards the pixels the experts argued over.

That per-pixel view is also the main limitation. Two experts drawing one mask and two drawing a clearly different one leave the same trace in D^{GT} as four experts disagreeing pixel by pixel, so the supervision cannot tell a disagreement about *which* lesion from a disagreement about *where its edge falls*, and neither can the metric we score it with. Beyond that, we trained for 100k of the paper's 240k iterations and used only the random crop for augmentation, so our absolute energy distances sit above the published ones; we tested one dataset and one anatomy; D^{GT} is estimated from just four experts; and each arm was run once with a single seed, so the smaller differences between arms, including arm B's lower Dice, cannot be separated from run-to-run variation.

Two extensions follow naturally. Restricting the alignment loss to a band around the predicted boundary would let it shape where edges are drawn without demanding uncertainty across whole lesions, the most likely route to a clearer Q2 result. A comparison against the Hierarchical Probabilistic U-Net, a follow-up by the same authors that models ambiguity at several spatial scales instead of through one global latent, would show whether a hierarchical latent handles existence-versus-boundary disagreement better than our single six-dimensional z.

References

[1] S. A. A. Kohl et al., *A Probabilistic U-Net for Segmentation of Ambiguous Images*, NeurIPS 2018. arXiv:1806.05034.

[2] S. A. A. Kohl et al., *A Hierarchical Probabilistic U-Net for Modeling Multi-Scale Ambiguity*, 2019. arXiv:1905.13077. (Source of the preprocessed LIDC crops.)

[3] S. G. Armato III et al., *The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI)*, Medical Physics, 2011.

[4] O. Ronneberger, P. Fischer, T. Brox, *U-Net: Convolutional Networks for Biomedical Image Segmentation*, MICCAI 2015.

[5] L. Zhang et al., *Disentangling Human Error from the Ground Truth in Segmentation of Medical Images*, NeurIPS 2020.